

Air Freight Premium

USAID health commodity shipments · 10,324 deliveries · 43 countries · 2006–2015

Year

All

Mode

All

Route

All

Group

All



10,324

Shipments

\$68.8M

Freight Spend

\$1.6bn

Commodity Value

5.79%

Freight Intensity

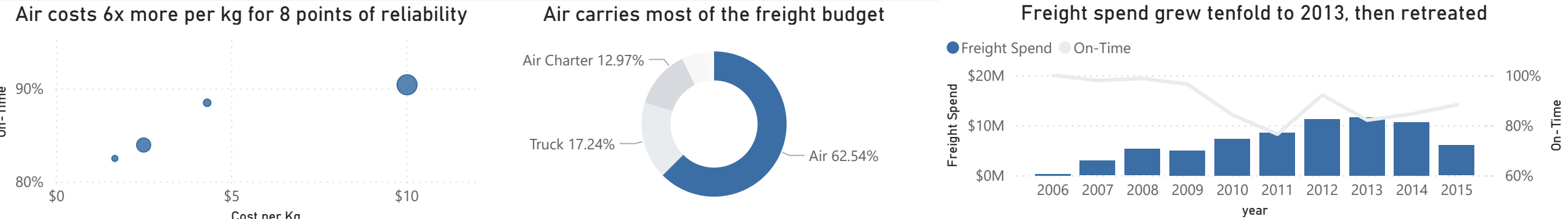
88.5%

On-Time

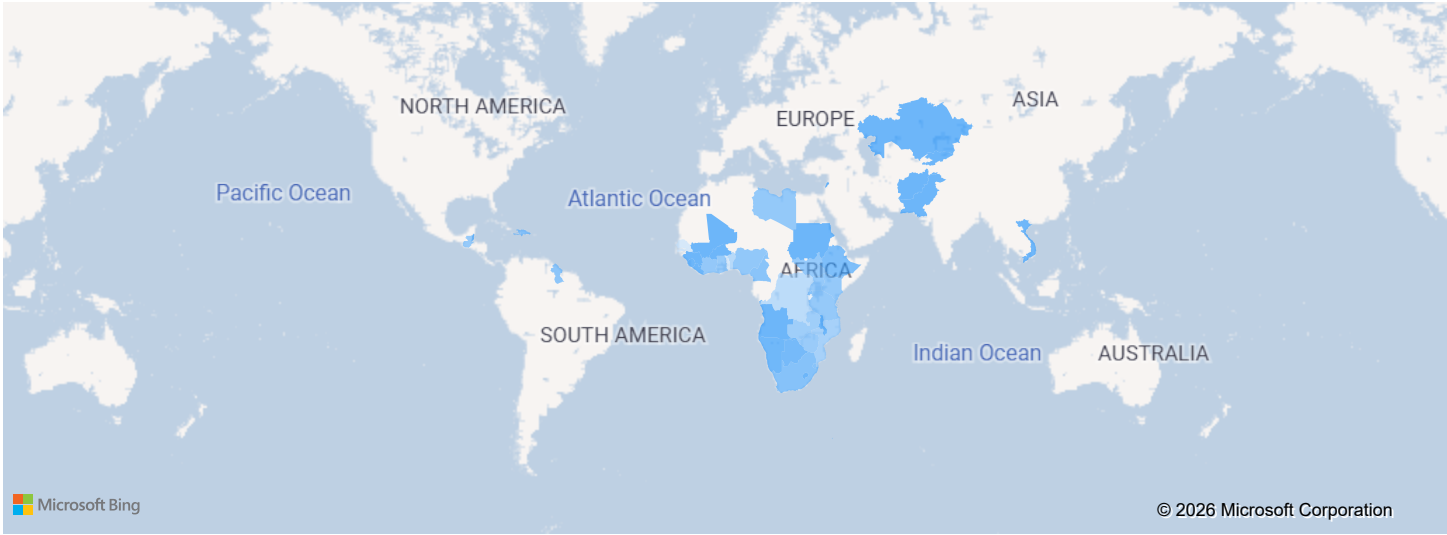
\$7.26

Cost per Kg

Air freight costs 6.0 times as much per kilogram as ocean, and buys 7.9 percentage points of on-time performance.



Eight of the ten largest destinations are in sub-Saharan Africa



Nigeria absorbs \$14.3M, more than double any other country

Country	Shipments	Freight Spend	On-Time	Cost per Kg
Nigeria	1,194	\$14.27M	88.1%	\$7.82
Zambia	683	\$6.31M	84.2%	\$4.00
Côte d'Ivoire	1,083	\$6.09M	87.6%	\$6.80
Rwanda	430	\$5.89M	91.4%	\$9.95
Uganda	779	\$5.48M	87.4%	\$7.76
Tanzania	519	\$4.83M	87.3%	\$5.20
Zimbabwe	538	\$4.32M	85.7%	\$4.82
Haiti	655	\$4.00M	90.5%	\$11.49
Mozambique	631	\$3.29M	81.6%	\$2.47
Vietnam	688	\$2.74M	99.1%	\$5.12
South Africa	1,406	\$2.49M	91.8%	\$1.41
Total	9,667	\$64.59M	88.5%	\$6.84

Freight cost reported for 60.0% of shipments; 23.7% were consolidated into other shipments. Cost measures cover reported shipments only. Source: USAID SCMS delivery history.

Delivery Reliability

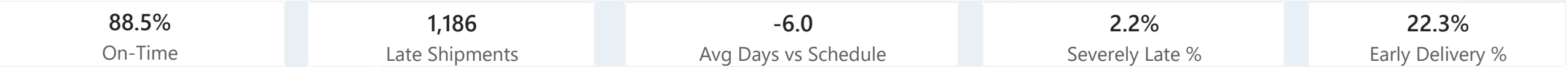
Where the programme misses its own delivery dates

Year
All

Mode
All

Route
All

Group
All

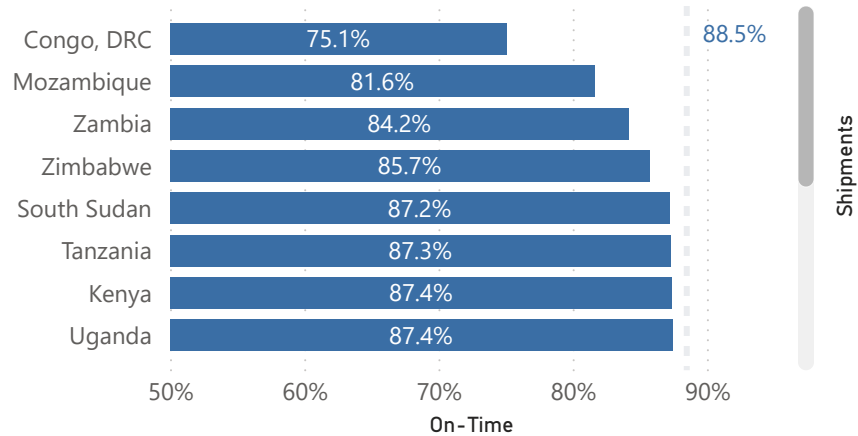


Show me
On-Time

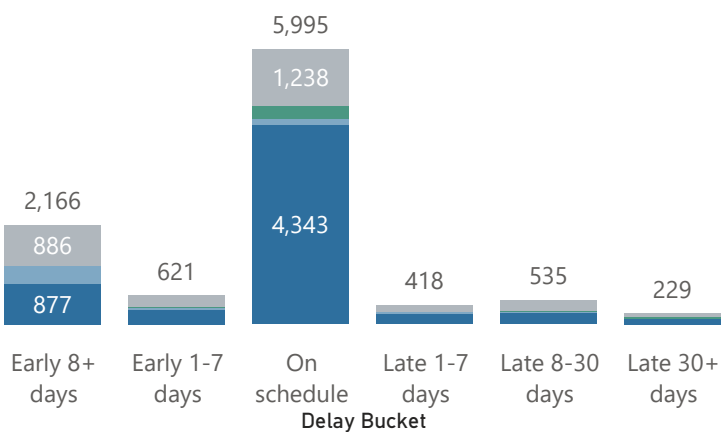
Break down by
Country

Weakest destination: Congo, DRC at 75.1% on-time across 333 shipments. Programme average is 88.5%.

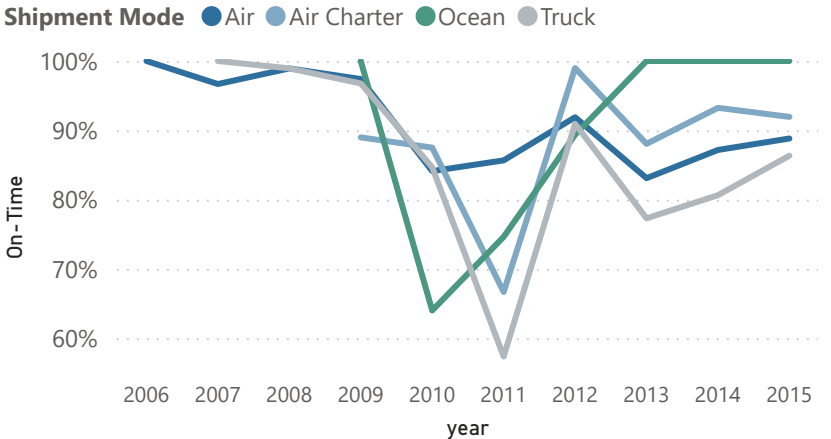
Weakest performers — use the selectors above



Most shipments land exactly on schedule



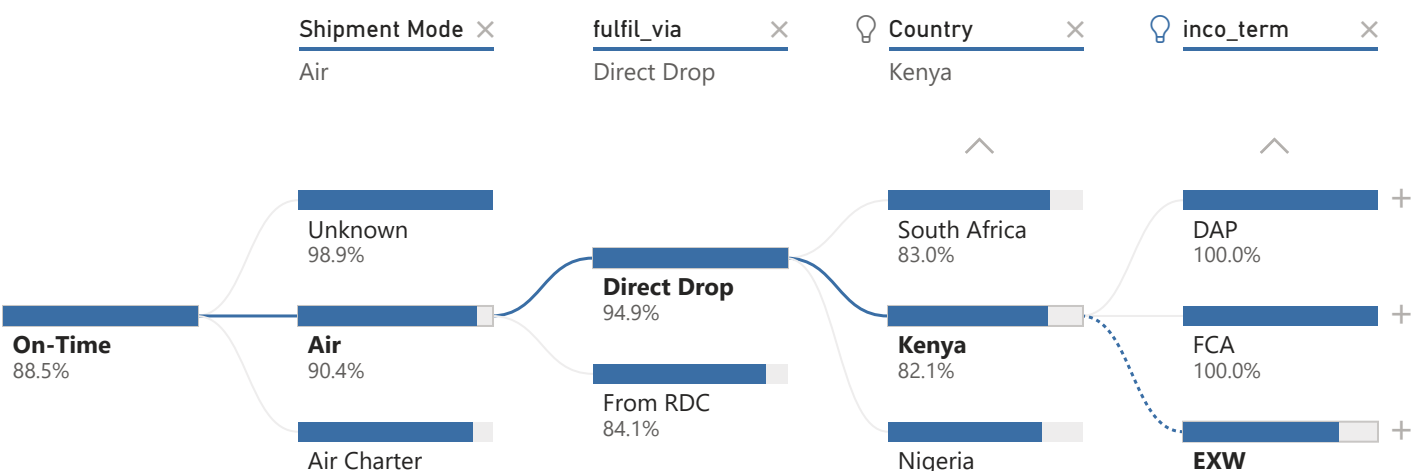
Reliability by mode over time



Reliability varies more by destination than by mode

Country	Air	Air Charter	Ocean	Truck	Total
Mozambique	96.9%			68.2%	81.6%
Côte d'Ivoire	89.6%			77.7%	86.3%
Uganda	89.7%			81.2%	87.3%
Nigeria	87.0%	88.3%			87.7%
Haiti	89.1%		100.0%		90.5%
South Africa	83.0%		71.6%	98.9%	91.6%
Vietnam	99.1%				99.1%
Total	91.2%	88.5%	80.3%	87.3%	89.2%

What drives lateness — click to explore



Country views limited to 16 destinations with 100+ shipments; matrix cells under 20 shipments suppressed. 2006 volume is ~4% of 2014, so early-year swings are noise. Source: USAID SCMS.

Freight Cost Analysis

What the programme pays to move goods, and why

Year

▼

All

▼

Mode

▼

All

▼

Route

▼

All

▼

Group

▼

All

▼

\$68.8M Freight Spend	\$7.26 Cost per Kg	5.79% Freight Intensity	\$2.41M Insurance Spend	3,424 Avg Shipment Weight (Kg)	60.0% Freight Coverage %
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Median freight rate is \$7.26 per kg — ocean at \$1.68 against air at \$10.02. Freight is separately invoiced for only 60% of shipments; the rest is bundled into commodity cost.

Air costs 6x ocean per kilogram

Mode	Cost per Kg
Air	\$10.02
Air Charter	\$4.00
Truck	\$2.00
Ocean	\$1.68

Small shipments pay a heavy per-kilo premium

Weight Range (kg)	Cost per Kg
Under 100	\$30.63
100-499	\$11.49
500-1,999	\$7.12
2,000-9,999	\$3.43
10,000 kg+	\$1.11

Freight visibility is set by the contract term, not by record-keeping

freight_cost_usd_status

- Cross-referenced to ...
- Included in com...
- Invoiced separ...
- Reported

INCOTerm	Cross-referenced to ...	Included in com...	Invoiced separ...	Reported
N/A - From RDC	25%	2%	73%	0%
EXW	15%	5%	80%	0%
DDP	2%	68%	30%	0%
FCA	5%	10%	85%	0%
CIP	5%	40%	55%	0%
DDU	0%	75%	25%	0%

Rates vary three-fold across vendors

Vendor	Shipments	Freight Spend	Cost per Kg	On-Time
SCMS from RDC	5,404	\$34.9M	\$5.05	82.8%
Orgenics, Ltd	754	\$9.8M	\$17.23	87.0%
Aurobindo Pharma Limited	668	\$5.0M	\$5.11	85.9%
MYLAN LABORATORIES LTD (FORMERLY ...	317	\$3.1M	\$5.40	99.4%
HETERO LABS LIMITED	277	\$2.6M	\$4.05	99.3%
SHANGHAI KEHUA BIOENGINEERING CO...	70	\$2.4M	\$12.11	97.1%
Trinity Biotech, Plc	356	\$2.3M	\$15.04	99.7%
ABBVIE LOGISTICS (FORMERLY ABBOTT L...	347	\$2.0M	\$11.53	98.8%
CIPLA LIMITED	175	\$1.9M	\$4.02	86.9%
Standard Diagnostics, Inc.	98	\$1.3M	\$10.58	95.9%
Total	8,466	\$65.2M	\$7.05	86.3%

Freight consumes 36% of commodity value in South Sudan

Country	Freight Intensity
South Sudan	36%
Congo, DRC	25%
Haiti	15%
Guyana	14%
Ethiopia	12%
Rwanda	10%
Uganda	8%
Côte d'Ivoire	8%
Zimbabwe	6%

Sajith Pamarathna

[View On GitHub](#)

Rates are medians, not averages: freight costs span \$0.75 to \$290,000. Cost measures cover the 60.0% of shipments with separately invoiced freight; weight is recorded for 61.7%. Source: USAID SCMS delivery history.

The Mode Shift Case

What switching non-urgent air freight to ocean would release

Year

All

Mode

All

Route

All

Group

All



Minimum days of schedule slack

30

244

Shiftable Shipments

1.15M

Shiftable Weight (Kg)

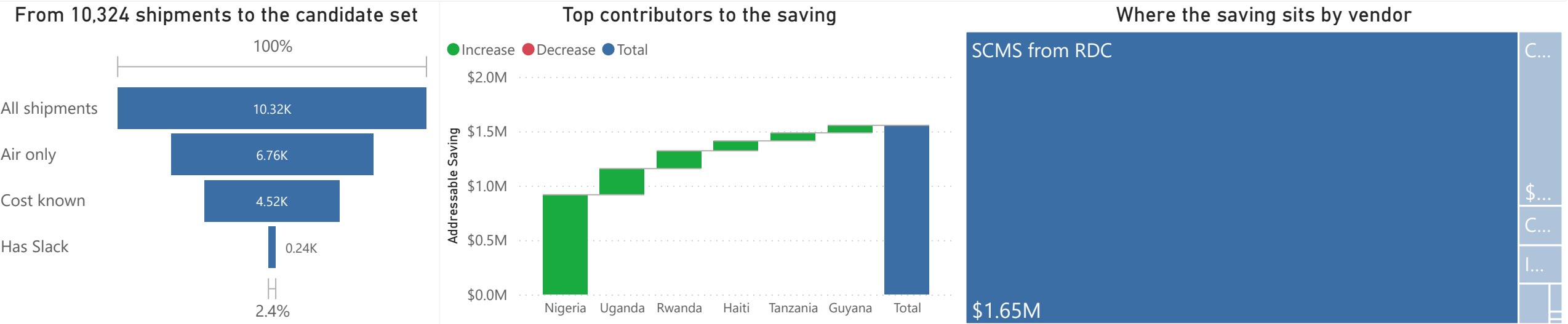
\$1.79M

Addressable Saving

3.44%

Saving as % of Air Spend

At 30+ days of schedule slack, 244 air shipments carrying 1,146,985 kg could have moved by ocean, releasing \$1,785,202 — 3.4% of air freight spend.



Highest-value shift candidates					
Country	Product Group	Shiftable Shipments	Shiftable Weight	Addressable Saving	Avg Days vs Schedule
Nigeria	ARV	97	957.00K	\$847.40K	-13.8
Uganda	ARV	30	48.30K	\$240.15K	-8.0
Rwanda	ARV	19	30.28K	\$150.85K	-6.8
Haiti	ARV	10	6.53K	\$75.18K	-2.6
Guyana	ARV	21	5.43K	\$69.34K	-25.5
Côte d'Ivoire	ARV	17	15.98K	\$66.05K	-6.4
Total		194	1,063.52K	\$1,448.97K	-9.0

- Recommended actions
1. Consolidate small shipments.

Freight costs \$30.63/kg under 100 kg against \$1.11/kg above 10,000 kg, a 27-fold spread. Batching is worth more than rate negotiation and is fully within the programme's control.
2. Re-baseline the air default on proven candidates.

244 shipments arrived 30+ days early yet moved by air. Shifting only those releases \$1.79M, 3.4% of air freight spend. Start with the evidenced cases rather than a blanket policy change.
3. Review the RDC route.

Direct drop delivers 94.9% on-time against 84.1% from the regional DC, at comparable cost. The cheaper route is also the less reliable one.

Modelled saving: median ocean rate (\$1.68/kg) applied to air shipments arriving 30+ days early. An opportunity estimate, not a forecast. Source: USAID SCMS.